Clinton Power Station
8401 Power Road
Clinton, IL 61727

U-604648

10 CFR 50.73
SRRS 5A.108

August 19, 2021

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D.C. 20555-0001

Clinton Power Station, Unit 1
Facility Operating License No. NPF-62
NRC Docket No. 50-461

Subject: Licensee Event Report 2021-001-00

Enclosed is Licensee Event Report (LER) 2021-001-00: Core Monitoring System Software Modeling Error Results in Condition Prohibited by Technical Specifications. This report is being submitted in accordance with the requirements of 10 CFR 50.73.

There are no regulatory commitments contained in this report.

Should you have any questions concerning this report, please contact Mr. Dale Shelton, Regulatory Assurance Manager, at (217) 937-2800.

Respectfully,

Thomas D. Chalmers
Site Vice President
Clinton Power Station

Attachment: Licensee Event Report 2021-001-00

cc:

Regional Administrator - Region III
NRC Senior Resident Inspector - Clinton Power Station
Office of Nuclear Facility Safety - Illinois Emergency Management Agency

NRC FORM 366 (08-2020) U.S. NUCLEAR REGULATORY COMMISSION

APPROVED BY OMB: NO. 3150-0104 EXPIRES: 08/31/2023

LICENSEE EVENT REPORT (LER)

(See Page 3 for required number of digits/characters for each block)
(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name	2. Docket Number	3. Page
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4. Title
Core Monitoring System Software Modeling Error Results in Condition Prohibited by Technical Specifications

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number 05000
07	20	2021	2021	- 001 -	00	08	19	2021	Facility Name	Docket Number 05000

9. Operating Mode	10. Power Level
1	098

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	**10 CFR Part 73**
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(4)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(i)	**10 CFR Part 21**	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(1)(i)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(iii)	**10 CFR Part 50**	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.77(a)(2)(ii)
☐ 20.2203(a)(2)(iv)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	
☐ 20.2203(a)(2)(v)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	

☐ **OTHER** (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
William Brockschmidt, Reactor Engineer	(217) 937-3888

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)				

16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

On 07/20/21, with Clinton Power Station (CPS) operating at 98 percent power, it was determined CPS had operated in conditions prohibited by Technical Specifications (TS). This determination was based on a review of operational history and General Electric - Hitachi (GEH) Safety Communication SC 21-04, Revision 1, “Fuel Support Side Entry Orifice Meta-Stable Flow for 2 Beam Locations in the BWR/6 Reactors.” The review concluded there were periods on 5/25/19, and between 6/22/19 to 8/3/19, when TS 3.2.2, “Power Distribution Limits – Minimum Critical Power Ratio (MCPR)” was not met due to a Core Monitoring System software modeling error. The error was identified by GEH and was caused by underprediction of Side Entry Orifice (SEO) loss coefficient for some fuel bundle locations which impacted the Maximum Fraction of the Limiting Critical Power Ratio (MFLCPR). The current MFLCPR value was verified within the recommended limit. Operations Standing Orders were issued to capture recommended actions from SC 21-04. The Core Operating Limit Report and Databank Design Analysis were revised and the core monitoring system was updated to address SC 21-04. This condition is reportable under 10 CFR 50.73(a)(2)(i)(B), any operation or condition which was prohibited by the plant's TSs. There was no impact to the health and safety of the public or plant personnel because the Safety Limit MCPR was not exceeded in previous cycles and the available margin was sufficient to accommodate the penalty evaluated in SC 21-04.

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		YEAR	SEQUENTIAL NUMBER	REV NO.
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NARRATIVE

PLANT AND SYSTEM IDENTIFICATION

General Electric -- Boiling Water Reactor, 3473 Megawatts Thermal Rated Core Power
Energy Industry Identification System (EIIS) codes are identified in text as [XX].

EVENT IDENTIFICATION

Core Monitoring System Software Modeling Error Results in Condition Prohibited by Technical Specifications

A. Plant Operating Conditions Before the Event

Unit: 1	Event Date: July 20, 2021	Event Time: N/A
Mode: 1	Mode Name: Power Operation	Reactor Power: 098

B. Description of Event

Minimum Critical Power Ratio (MCPR) is a ratio of the fuel assembly power that would result in the onset of boiling transition to the actual fuel assembly power. The Operating Limit MCPR (OLMCPR) is established to ensure that no fuel damage results during anticipated operational occurrences, and that 99.9% of the fuel rods avoid boiling transition if the limit is not violated. Although fuel damage does not necessarily occur if a fuel rod experiences boiling transition, the critical power at which boiling transition is calculated to occur has been adopted as a fuel design criterion. The core monitoring system applies the appropriate OLMCPR limits incorporating the appropriate flow dependent MCPR (MCPRF) or the power dependent MCPR (MCPRP) limits; maximum of the two off-rated limits is applied for added conservatism. When this OLMCPR is divided by the calculated actual MCPR, this returns the Maximum Fraction of the Limiting Critical Power Ratio (MFLCPR). With MFLCPR <1.000, the Core Operating Limits Report (COLR) limits are met such that Technical Specification (TS) 3.2.2, Power Distribution Limits – Minimum Critical Power Ratio (MCPR), is satisfied. With MFLCPR > 1.000, TS 3.2.2 requires MFLCPR to be restored within 2 hours or power reduced to below 21.6% Rated Thermal Power within the next 4 hours.

On July 20, 2021, it was determined that a core monitoring system software modeling error resulted in multiple entries into a condition prohibited by TS. This determination was reached after detailed review of plant operational history and information provided in General Electric - Hitachi (GEH) Safety Communication (SC) 21-04, "Fuel Support Side Entry Orifice Meta-Stable Flow for 2 Beam Locations in the BWR/6 Reactors," Revision 1, issued on June 17, 2021.

SC 21-04, Revision 0, issued on April 19, 2021, identified a flow pattern that may periodically form under certain core conditions in which fuel bundle bypass flow is not accurately modeled by the core monitoring system. This condition is applicable to BWR/6 stations utilizing GEH core

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monitoring software, which includes Clinton Power Station (CPS). The condition applies to fuel assembly locations which are adjacent to two supporting beams of the core plate. These locations represent approximately 25% of the fuel bundles in the CPS core. The condition identified in SC 21-04, Revision 0, impacts the limits specified in TS 3.2.2.

In response to SC 21-04, Revision 0, CPS issued Standing Order 2021-04, "MFLCPR Administrative Limit," which implemented a core wide penalty on the MFLCPR thermal limit. The Standing Order was based on recommended actions from SC 21-04, Revision 0.

On June 17, 2021, GEH revised SC 21-04. SC 21-04 (i.e., Revision 1) provided additional guidance and clarification on the application of the MFLCPR penalty recommended in Revision 0. Previously, GEH recommended a 0.05 penalty be placed on core wide MFLCPR. Upon further investigation SC 21-04, Revision 1, recommended the MFLCPR penalty be calculated via a plant specific formula, which results in differing penalty values for different core power, flow, and equipment out of service conditions. The full description of the recommended MFLCPR penalty is contained in Attachment 5 of SC 21-04, Revision 1. CPS Standing Order 2021-04 was revised to incorporate the new recommendations in SC 21-04, Revision 1, to ensure compliance with TS 3.2.2.

Based on information provided in SC 21-04, Revision 1, and a review of unit operational history, it has been determined there were periods on May 25, 2019, and between June 22 - August 3, 2019, when TS 3.2.2 was not met because, after implementation of SC 21-04, Revision 1, penalty, MFLCPR was greater than 1.0 for longer than 6 hours.

C. Cause of the Event

As stated in SC 21-04, Revision 1, GEH notified CPS the Side Entry Orifice (SEO) loss coefficient was underpredicted for some fuel bundle locations, which could result in an overprediction of MCPR margin in core monitoring applications. The overprediction was a result of not originally including in the assessment flow area restrictions associated with instrument support structures in the cross beams (structural supports underneath the core plate) in BWR/6 plant designs. The issue was identified in evaluations completed as part of follow-on actions from a 2020 Part 21 report (Non-Conservative BWR/6 Side Entry Orifice (SEO) Loss Coefficients, dated June 24, 2020, ML20176A432). This condition, the level of details used in vendor calculations, and the basis of the data used in the applied application were beyond the control of Exelon to identify, predict or prevent because it is a legacy vendor design calculation error. Further, Exelon does not have the capability to check these calculations, nor the resources to perform an independent review of GEH calculations.

D. Safety Consequences

The condition described in this LER is reportable under 10 CFR 50.73(a)(2)(i)(B), any operation or condition which was prohibited by the plant's TS. The OLMCPR is set such that if the plant is operating at the OLMCPR, the most limiting operational transient for the cycle will not result in violation of the Safety Limit MCPR (SLMCPR). Exelon reviewed the unit operating history for

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limiting transients and other analyzed transients that are close to being limiting in the Supplemental Reload Licensing Reports. Exelon concluded that, for the period reviewed, the limiting transients were turbine trip without bypass, feedwater controller failure full open, load reject without bypass, rod withdrawal error, loss of 100 degree Fahrenheit feedwater heating, and fuel loading/orientation error. Because these limiting or near limiting transients did not occur in the period reviewed, the margin between the SLMCPR and OLMCPR was adequate for continued SLMCPR protection. Therefore, the SLMCPR was not exceeded in previous cycles and the margin between SLMCPR and OLMCPR was sufficient to accommodate the penalty evaluated in SC 21-04, Revisions 0 and 1. As a result, there was no impact to the health and safety of the public or plant personnel from this condition. In addition, this event does not meet the criteria for a Safety System Functional Failure.

E. Corrective Actions

- Verified current MFLCPR value is within recommended limit.
- Operations Standing Orders issued to capture recommended actions from SC 21-04, Revisions 0 and 1.
- Revised the Core Operating Limit Report and Databank Design Analysis as necessary to address SC 21-04, Revision 1.
- Updated the core monitoring system to address SC 21-04, Revision 1.

F. Previous Similar Occurrences

A review of previous LERs did not identify any events that were similar to the condition described in this LER.

G. Component Failure Data

Not applicable to this event.